AWS CLOUD INFRASTRUCTURE & NETWORKING LAB REPORT

Multi-VPC Peering Architecture with Bastion SSH ProxyJump

Student Name: Himesh Kumar Yadav(Batch-1)	**AWS Region:** Asia Pacific (Sydney) [ap-southeast-2]
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1. Executive Summary & Architecture Overview

The objective of this engineering implementation was to design, construct, validate, and decommission a production-grade multi-VPC cloud networking infrastructure within the AWS Sydney region (ap-southeast-2). The primary architectural goal was to enable bi-directional, high-throughput, and private inter-workload communication between isolated application tiers without exposing the backend compute nodes to the public internet.

To eliminate internet-facing attack vectors, backend compute instances (ewb-app-a and ewb-app-b) reside within isolated private subnets lacking Internet Gateways or auto-assigned public IP addresses. Secure administrative ingress is achieved via a dedicated public-facing Bastion host (ewb-bastion) located in VPC A. Network operators traverse this perimeter using OpenSSH ProxyJump encapsulation over Port 22, ensuring that cryptographic private keys remain strictly on the local administration client and are never loaded onto intermediate cloud jump boxes.

Architecture Topology Breakdown

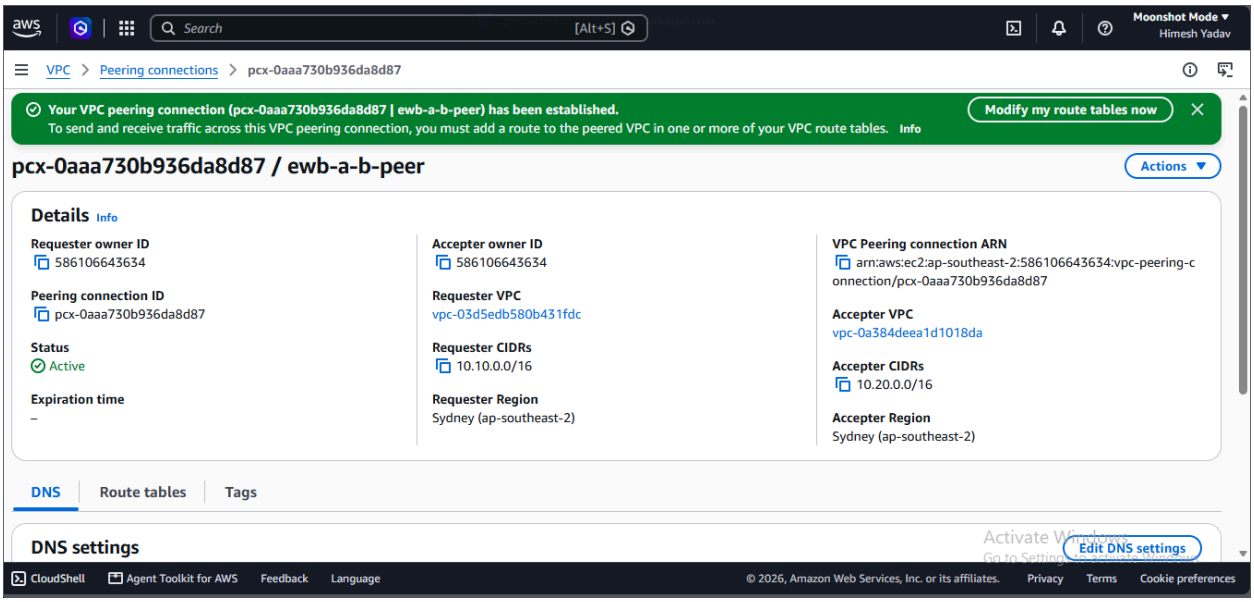
- **VPC A (CIDR: 10.10.0.0/16):** Dual-tier networking containing a Public Subnet (10.10.1.0/24) with an Internet Gateway (ewb-igw-a) attachment for Bastion ingress, and an isolated Private Subnet (10.10.2.0/24) hosting workload ewb-app-a.
- **VPC B (CIDR: 10.20.0.0/16):** Single-tier workload network containing an isolated Private Subnet (10.20.1.0/24) hosting workload ewb-app-b. It possesses zero Internet Gateways and no direct public ingress/egress path.
- **Inter-VPC Interconnect:** AWS VPC Peering Connection (pcx-0aaa730b936da8d87) establishing private, hardware-accelerated routing across the AWS global network backbone, maintaining sub-millisecond round-trip packet delivery.

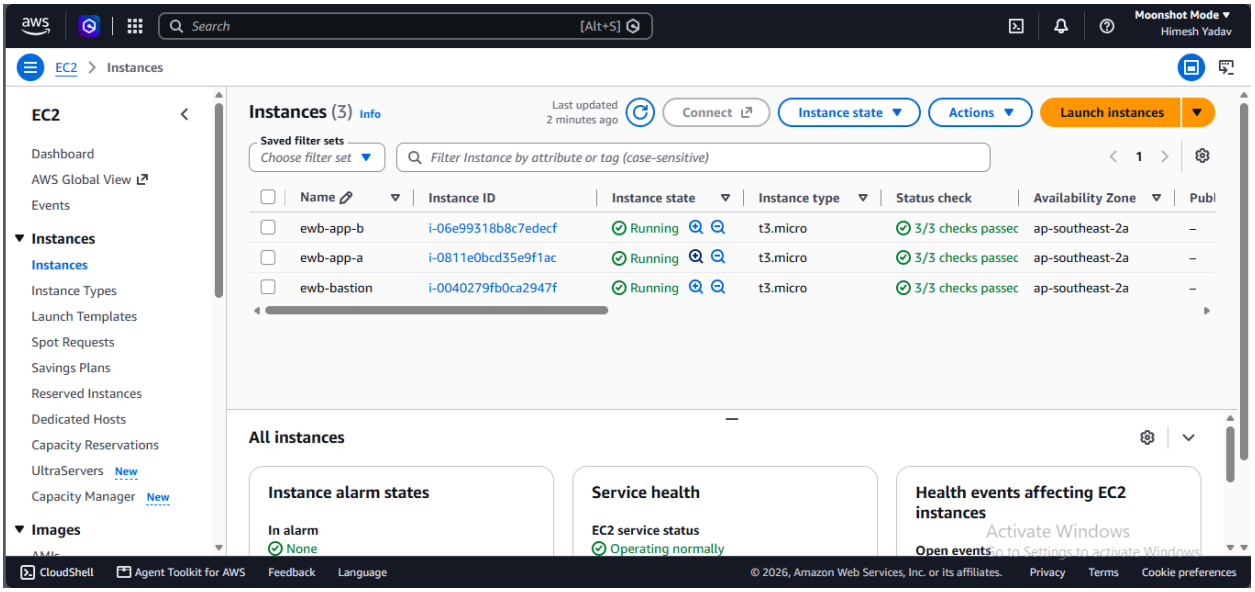
2. Comprehensive Resource Inventory

The following configuration matrix details all provisioned cloud infrastructure components, exact AWS resource identifiers, subnet allocations, and firewall attachments:

Resource Name	AWS Identifier	Type / Subnet	CIDR / Primary IP	Security Group
ewb-vpc-a	vpc-03d5edb580b431fdc	VPC (Requester)	10.10.0.0/16	N/A (Root Network)
ewb-vpc-b	vpc-0a384deea1d1018da	VPC (Accepter)	10.20.0.0/16	N/A (Root Network)
ewb-a-public	subnet-0d10a8805db1b08b8	Public Subnet (VPC A)	10.10.1.0/24	Associated to ewb-rt-a-public

ewb-a-private	subnet-0f25db438e03130fa	Private Subnet (VPC A)	10.10.2.0/24	Associated to ewb-rt-a-private
ewb-b-private	subnet-0b6549350c362aaf1	Private Subnet (VPC B)	10.20.1.0/24	Associated to ewb-rt-b-private
ewb-igw-a	igw-08392545212ebd105	Internet Gateway	0.0.0.0/0 Target	Attached to VPC A
ewb-rt-a-public	rtb-0a0680a6...	Public Route Table	0.0.0.0/0 -> IGW	pcx-0aaa730b936da8d87 target
ewb-rt-a-private	rtb-06bc585fae14ceb80	Private Route Table	10.20.0.0/16 -> PCX	pcx-0aaa730b936da8d87 target
ewb-rt-b-private	rtb-08b90a5ddcb47db33	Private Route Table	10.10.0.0/16 -> PCX	pcx-0aaa730b936da8d87 target
ewb-bastion	i-0040279fb0ca2947f	EC2 (t3.micro)	10.10.1.10 (Static EIP)	ewb-sg-bastion (sg-0c82ca090c31da21d)
ewb-app-a	i-0811e0bcd35e9f1ac	EC2 (t3.micro)	10.10.2.10 (No Public IP)	ewb-sg-app-a (sg-01311ba...)
ewb-app-b	i-06e99318b8c7edecf	EC2 (t3.micro)	10.20.1.10 (No Public IP)	ewb-sg-app-b (sg-090e977209ed3956d)
ewb-a-b-peer	pcx-0aaa730b936da8d87	VPC Peering Connection	VPC A <--> VPC B	Status: Active (Accepted)





3. Security Posture & SSH ProxyJump Architecture

The security model strictly enforces least-privilege perimeter isolation through defense-in-depth security group configurations and local client key management:

Security Group Rules Definition

- **ewb-sg-bastion (Perimeter Group):** Inbound Port 22 (SSH) strictly restricted to administrator's specific external public IP (/32). Default outbound all traffic permitted.
- **ewb-sg-app-a (Workload A Group):** Inbound Port 22 restricted exclusively to Bastion internal IP (10.10.1.10/32). Inbound Custom ICMP (Echo Request) permitted exclusively from workload ewb-app-b (10.20.1.10/32).
- **ewb-sg-app-b (Workload B Group):** Inbound Port 22 restricted exclusively to Bastion internal IP (10.10.1.10/32). Inbound Custom ICMP (Echo Request) permitted exclusively from workload ewb-app-a (10.10.2.10/32).

Client-Side OpenSSH ProxyJump Configuration (~/.ssh/config)

Using OpenSSH ProxyJump, the private key (ewb-lab.pem) remains exclusively within the administrator's local workstation. The Bastion host acts purely as a transport relay without access to decrypted SSH keys:

```
Host ewb-bastion
    HostName 54.206.27.213   # Associated Elastic IP Address
    User ec2-user
    IdentityFile ~/.ssh/ewb-lab.pem
    IdentitiesOnly yes

Host ewb-app-a
    HostName 10.10.2.10
    User ec2-user
    IdentityFile ~/.ssh/ewb-lab.pem
    IdentitiesOnly yes
    ProxyJump ewb-bastion
```

```
Host ewb-app-b
    HostName 10.20.1.10
    User ec2-user
    IdentityFile ~/.ssh/ewb-lab.pem
    IdentitiesOnly yes
    ProxyJump ewb-bastion
```

4. Empirical Verification & Test Results

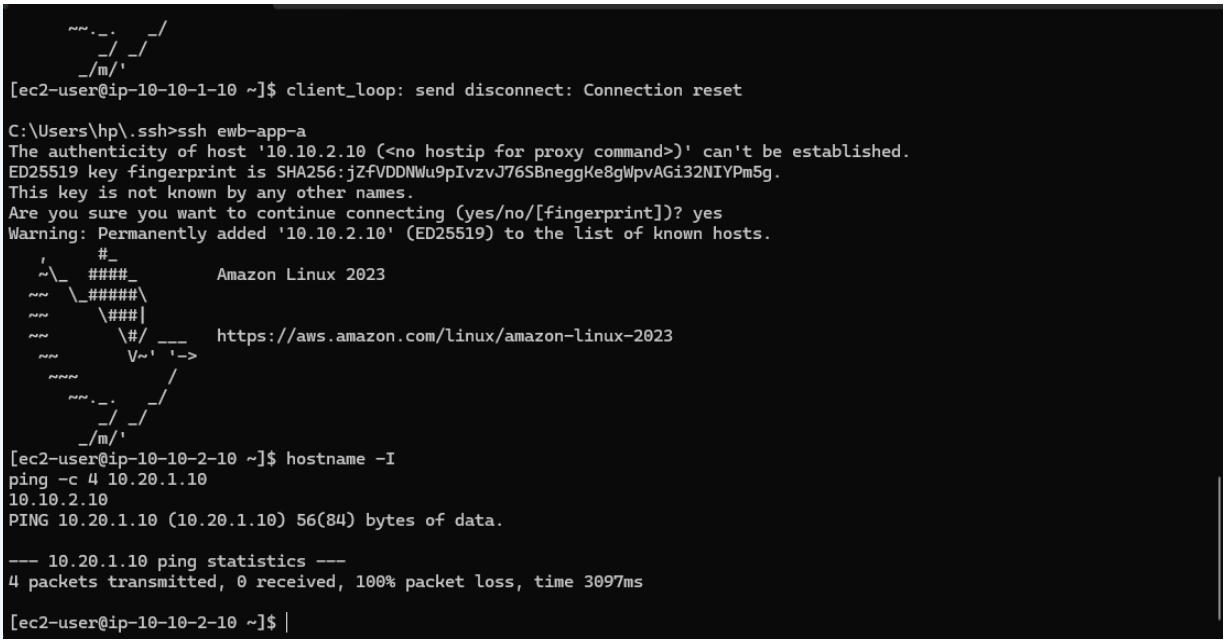
A rigorous four-phase testing protocol was conducted to demonstrate packet isolation, active peering propagation, bi-directional routing, and granular firewall enforcement:

Test 01: Baseline Negative Connectivity Test (Pre-Peering)

Before initiating the VPC Peering connection, ICMP packets were transmitted from ewb-app-a (10.10.2.10) targeting ewb-app-b (10.20.1.10). Without cross-VPC route propagation, 100% packet loss occurred as expected.

```
C:\Users\hp>ssh ewb-app-a "ping -c 4 -W 2 10.20.1.10"
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.

--- 10.20.1.10 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3099ms
[VERDICT: PASSED - Isolation Verified]
```

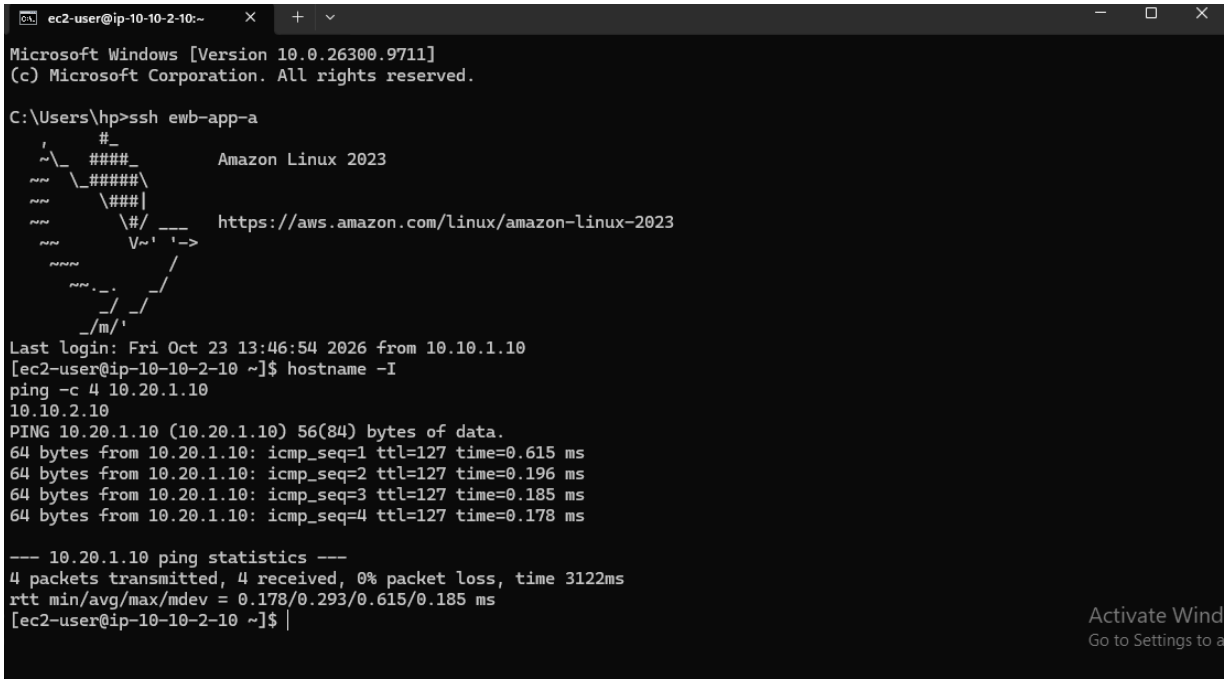


Test 02: Positive Cross-VPC ICMP Connectivity Test (Post-Peering)

Following VPC Peering establishment (pcx-0aaa730b936da8d87) and route updates across all route tables, the identical ping command was executed from ewb-app-a:

```
[ec2-user@ip-10-10-2-10 ~]$ hostname -I
10.10.2.10
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
64 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=0.615 ms
64 bytes from 10.20.1.10: icmp_seq=2 ttl=127 time=0.196 ms
64 bytes from 10.20.1.10: icmp_seq=3 ttl=127 time=0.185 ms
64 bytes from 10.20.1.10: icmp_seq=4 ttl=127 time=0.178 ms

--- 10.20.1.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3122ms
rtt min/avg/max/mdev = 0.178/0.293/0.615/0.185 ms
[VERDICT: PASSED - Low-latency Cross-VPC Peering Active]
```



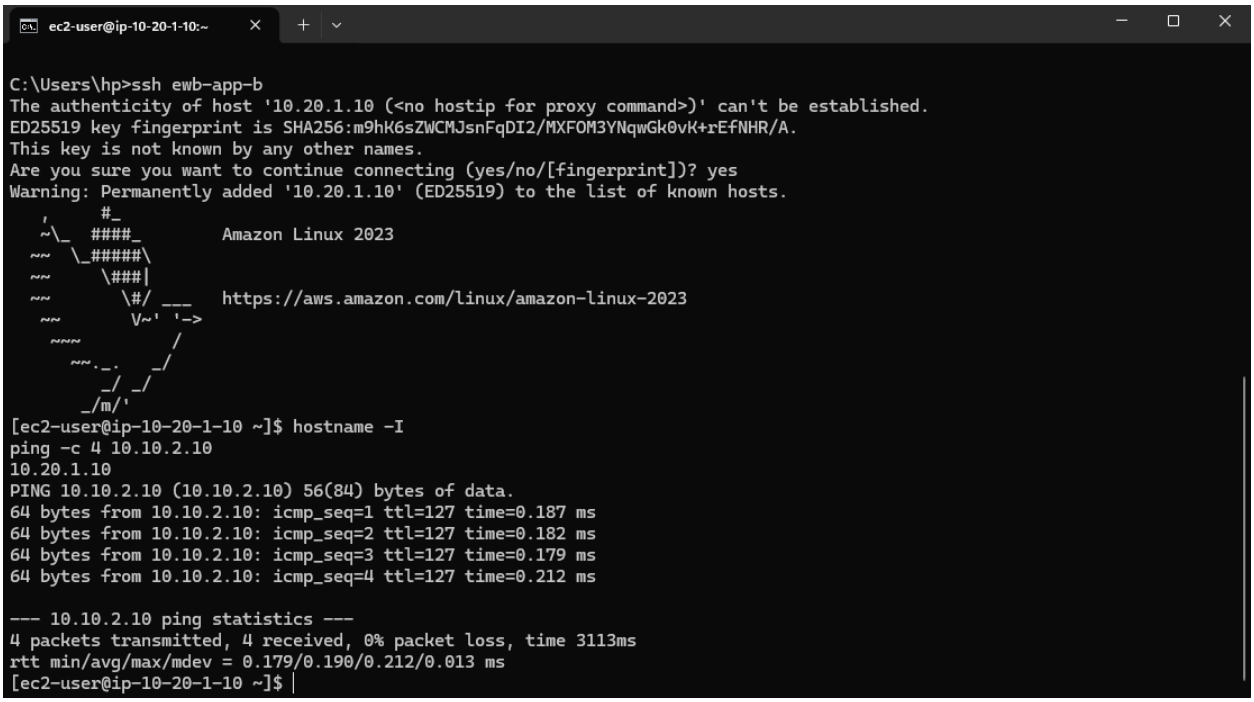
Test 03: Reverse Bi-Directional Path Verification

Connected directly into ewb-app-b (10.20.1.10) via SSH ProxyJump to verify reverse packet transit targeting ewb-app-a (10.10.2.10):

```
[ec2-user@ip-10-20-1-10 ~]$ hostname -I
10.20.1.10
[ec2-user@ip-10-20-1-10 ~]$ ping -c 4 10.10.2.10
```

```
PING 10.10.2.10 (10.10.2.10) 56(84) bytes of data.
64 bytes from 10.10.2.10: icmp_seq=1 ttl=127 time=0.182 ms
64 bytes from 10.10.2.10: icmp_seq=2 ttl=127 time=0.179 ms
64 bytes from 10.10.2.10: icmp_seq=3 ttl=127 time=0.190 ms
64 bytes from 10.10.2.10: icmp_seq=4 ttl=127 time=0.212 ms

--- 10.10.2.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3113ms
rtt min/avg/max/mdev = 0.179/0.190/0.212/0.013 ms
[VERDICT: PASSED - Symmetric Return Routing Confirmed]
```

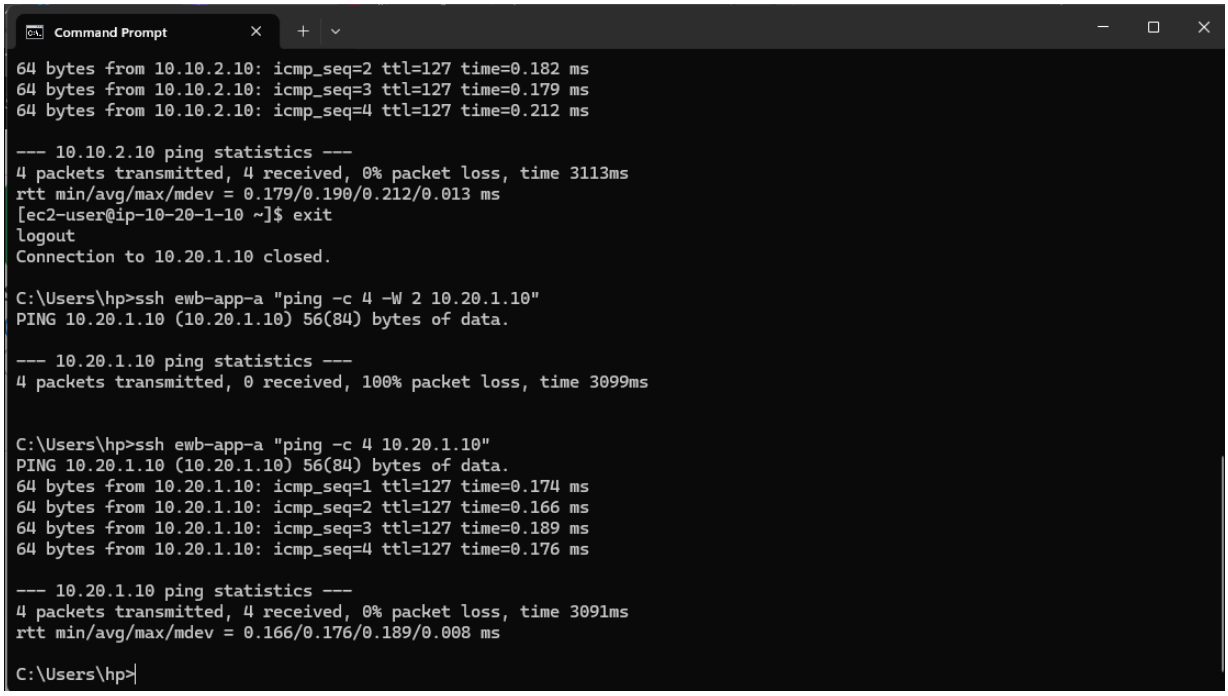


Test 04: Controlled Security Group Failure & Restoration Test

To confirm that packet transmission is actively filtered by instance security groups and not solely by route tables, the inbound ICMP rule on ewb-sg-app-b was removed. The ping immediately failed (100% loss). Upon re-adding the rule, connectivity was restored instantly (0% loss, ~0.176ms):

```
# Rule Removed in Console:
C:\Users\hp>ssh ewb-app-a "ping -c 4 -W 2 10.20.1.10"
--- 10.20.1.10 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3099ms

# Rule Restored in Console:
C:\Users\hp>ssh ewb-app-a "ping -c 4 10.20.1.10"
--- 10.20.1.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3091ms
rtt min/avg/max/mdev = 0.166/0.176/0.189/0.008 ms
[VERDICT: PASSED - Stateful Firewall Filtering Empirically Validated]
```



5. Technical Troubleshooting & Architectural Reflection

- **Regional vCPU Quota Management:** When deploying ewb-app-b, an AWS API error reported vCPU quota exhaustion (limit of 8). Analysis showed prior stopped laboratory instances occupying allocation. Terminating legacy sandbox machines freed up regional vCPUs immediately, allowing standard t3.micro deployment.
- **Dynamic Public IP vs. Elastic IP Persistence:** Stopping the Bastion instance overnight to pause lab billing released its dynamic public IPv4 address, causing 'Connection timed out' on subsequent attempts. This was resolved by allocating an AWS Elastic IP (EIP) and associating it persistently with ewb-bastion, keeping client SSH configs static across stop/start cycles.
- **Non-Transitive Peering Boundary:** AWS VPC Peering is non-transitive by design. While VPC A can communicate directly with VPC B, VPC B cannot route through VPC A's Internet Gateway to access the internet. This maintains security compliance for air-gapped backend database workloads in VPC B.

6. Resource Decommissioning & Zero-Cost Cleanup Log

Following full empirical validation, all deployed AWS cloud resources were systematically dismantled in reverse dependency order to prevent any ongoing billing charges:

- **1. EC2 Compute Layer:** Terminated ewb-bastion, ewb-app-a, and ewb-app-b instances; verified root EBS volumes terminated.
- **2. Elastic IP Address:** Disassociated and released Elastic IP back to AWS pool to avoid idle reservation fees.
- **3. VPC Peering Interconnect:** Deleted ewb-a-b-peer (pcx-0aaa730b936da8d87) and purged related route entries.
- **4. Network Perimeter & VPCs:** Detached and deleted ewb-igw-a, deleted custom security groups, and purged ewb-vpc-a and ewb-vpc-b along with child subnets and custom route tables.